Dynamical Systems

Part II

Recorded from the lecturer in 2026-2027.

Dynamical systems – mathematical entities described by a set of equations for a set of variables w.r.t. a time-like variable.

State space or phase space.

∗ Continuous time ⇒ set of ODEs

Lotka–Volterra for rabbits (r) & sheep (s)

$$\dot{r} = r(a - br - cs), \qquad \dot{s} = s(d - er - fs), \qquad a, b, \ldots, f > 0.$$

$$\text{State space} \qquad [0, \infty) \times [0, \infty).$$

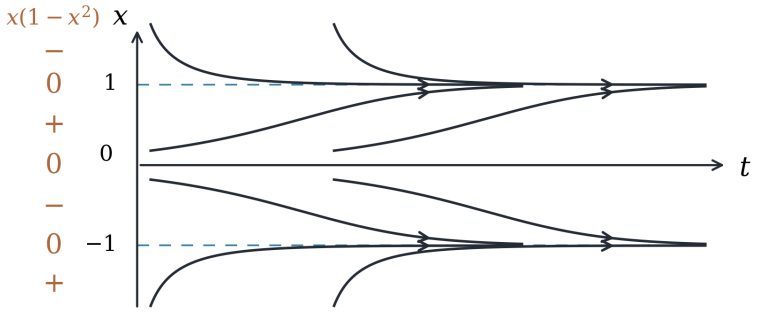
e.g. Logistic map $\qquad x_{n+1} = \mu x_n (1 - x_n), \quad 0 \le \mu \le 1$

Questions to address

- ∗ Long-term behavior of the system?
- ∗ Are there periodic solutions
- ∗ Stability of solutions, chaos
- ∗ Effect of system parameter variation

Example 1D system $\qquad \dot{x} = x(1 - x^2)$

$$\text{Solution} \qquad x(t) = \pm \left(1 + 2e^{2(t_0 - t)}\right)^{-1/2}$$



as $t \to \infty$

$x(0) > 0 \Rightarrow x(t) \to 1$

$x(0) < 0 \Rightarrow x(t) \to -1$

$x(0) = 0 \Rightarrow x(t) = 0$

Example 2D, Lotka–Volterra

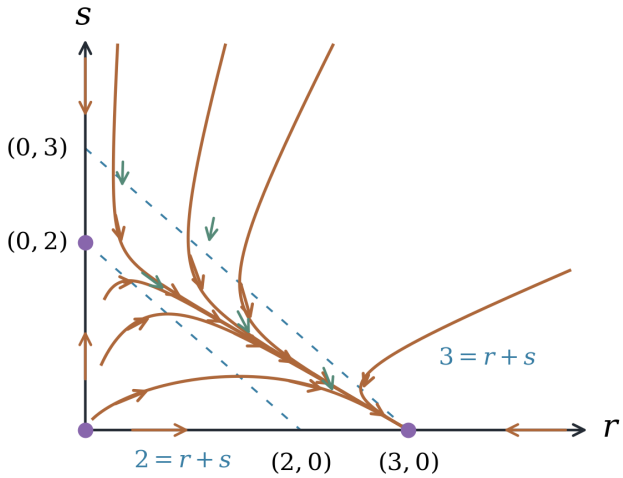
$$\dot{r} = r(3 - r - s)$$
$$\dot{s} = s(2 - r - s)$$

Can get $r(s) \Rightarrow t(s)$

∗ Find fixed points

$$\dot{r}, \dot{s} = 0 : \quad (0,0), (3,0), (0,2)$$

∗ Look at $\dot{r}, \dot{s}$ in state space



$$\text{As } t \to \infty \qquad (r,s) \to \begin{cases} (3,0), & r > 0, \\ (0,2), & r = 0,\ s > 0, \\ (0,0), & r = s = 0. \end{cases}$$

Example predator-prey

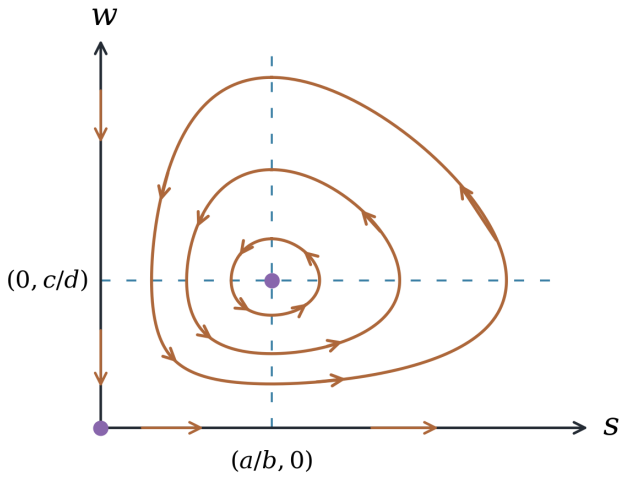
sheep (s) & wolves (w)

$$\dot{w} = w(-a + bs)$$
$$\dot{s} = s(c - dw)$$

$$a, \ldots, d > 0$$

Fixed points

$$(0,0), \quad (a/b, c/d)$$



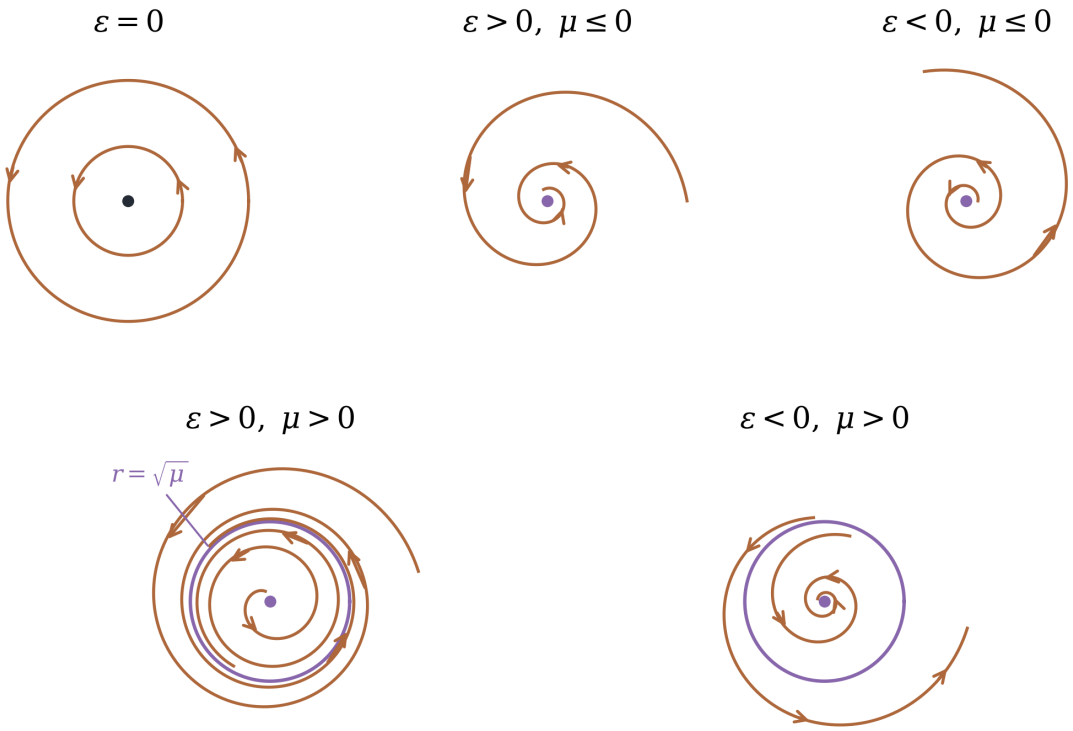
$$(s,w) \to \begin{cases} \text{periodic around } (a/b, c/d), & s > 0,\ w > 0, \\ (0,0), & s = 0,\ w \geq 0, \\ (\infty, 0), & s > 0,\ w = 0 \end{cases} \qquad \text{as } t \to \infty$$

Example

$$\dot{x} = -y + \varepsilon x(\mu - x^2 - y^2)$$
$$\dot{y} = \quad x + \varepsilon y(\mu - x^2 - y^2)$$

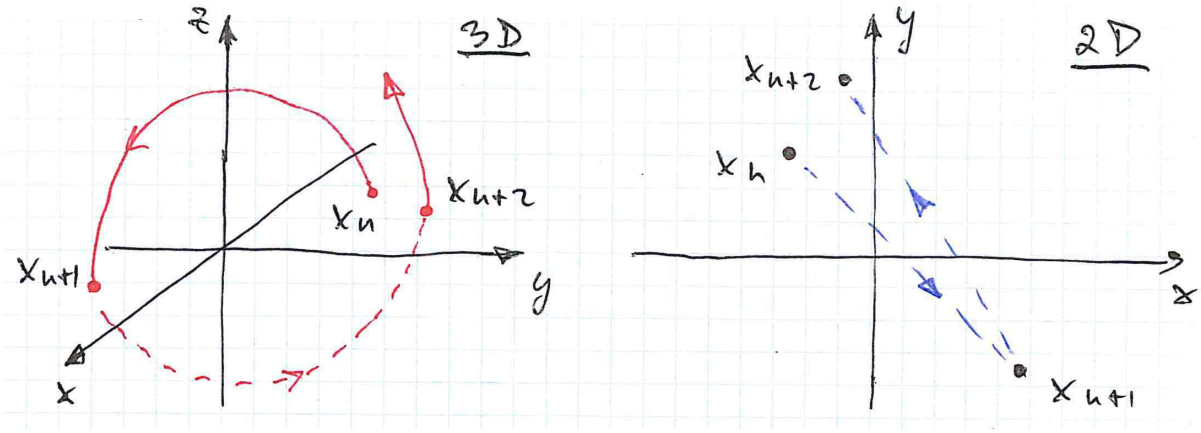
Switch to polars:

$$r = \sqrt{x^2 + y^2}, \qquad \theta = \operatorname{atan} \frac{y}{x},$$
$$\dot{r} = \frac{x\dot{x} + y\dot{y}}{r}, \qquad \dot{\theta} = \frac{x\dot{y} - y\dot{x}}{r^2} \quad \Rightarrow$$
$$\underline{\dot{r} = \varepsilon r(\mu - r^2)}, \qquad \underline{\dot{\theta} = 1}$$



End of lecture 1

Poincare map in 2D for a 3D dynamical system:



Can also consider maps as discretized ODEs.

Plan for the course

(3)	§1 :	Basic definitions	ODEs
(3)	§2 :	Fixed points	
(2)	§3 :	Stability	
(5)	§4 :	Periodic orbits	
(5)	§5 :	Bifurcations	
(3)	§6 :	Maps	Maps
(3)	§7 :	Chaos	

§1. Definitions

(1.1) Define a State Space (Phase Space) $E \subseteq \mathbb{R}^n$. The state of the system is described by $\bar{x} \in E$. The state evolves in time according to an ODE of the form

$$\dot{\bar{x}} = \frac{d\bar{x}}{dt} = \bar{f}(\bar{x}, t), \qquad \text{where } \bar{f} : E \times \mathbb{R} \to E \text{ is a vector field.}$$

If $\dfrac{\partial \bar{f}}{\partial t} = 0$ the system is autonomous.

A non-autonomous system reduces to an autonomous by defining $\bar{y} = (\bar{x}, t) \in E \times \mathbb{R}$ so that

$$\dot{\bar{y}} = \begin{pmatrix} \bar{f}(\bar{y}) \\ 1 \end{pmatrix}.$$

Thus, can study only autonomous systems.

A nth order ODE

$$\frac{d^n x}{dt^n} = g\left(x, \frac{dx}{dt}, \ldots, \frac{d^{n-1}x}{dt^{n-1}}\right)$$

can be reduced to a standard form with n equations by introducing

$$\bar{y} = \left(x, \frac{dx}{dt}, \ldots, \frac{d^{n-1}x}{dt^{n-1}}\right) = (y_1, y_2, \ldots, y_n)$$

so that

$$\dot{\bar{y}} = \begin{pmatrix} y_2 \\ y_3 \\ \vdots \\ y_n \\ g(y_1, y_2, \ldots, y_n) \end{pmatrix}.$$

Similarly to autonomous systems, will only consider maps of the form

$$\bar{x}_{n+1} = \bar{F}(\bar{x}_n) \qquad \text{and not} \qquad \bar{F}(\bar{x}_n, n).$$

1.2 Initial value problem (IVP)

We would like to explore an initial value problem for $\dot{\bar{x}} = \bar{f}(\bar{x}, t)$:

> Given the initial value of $\bar{x}$: $\bar{x}(t_0) = \bar{x}_0$, $\bar{x}_0 \in E$, $t \in I \subseteq \mathbb{R}$, find a differentiable function $\bar{x}(t)$ for $t \in I$ which remains in E for $t \in I$ and satisfies the initial condition and the ODE.

For an autonomous system can define solution in terms of a flow ϕ_t.

Definition (Flow) $\phi_t(\bar{x})$ such that $\phi_t(\bar{x}_0)$ is the solution of $\dot{\bar{x}} = \bar{f}(\bar{x})$ starting at $\bar{x}_0$ when $t = 0$ is called the flow through $\bar{x}_0$ at $t = 0$.

Thus,

$$\phi_0(\bar{x}_0) = \bar{x}_0, \qquad \phi_s\big(\phi_t(\bar{x}_0)\big) = \phi_{s+t}(\bar{x}_0).$$

Does this solution exist? Is it unique?

Theorem (Cauchy–Peano)

If $\bar{f}(\bar{x}, t)$ is continuous and $|\bar{f}| < M$ in the domain

$$D : \big\{|t - t_0| < \alpha,\ |\bar{x} - \bar{x}_0| < \beta\big\}$$

then the IVP has a solution for $|t - t_0| < \min(\alpha, \beta/M)$.

But uniqueness requires stronger conditions on f.

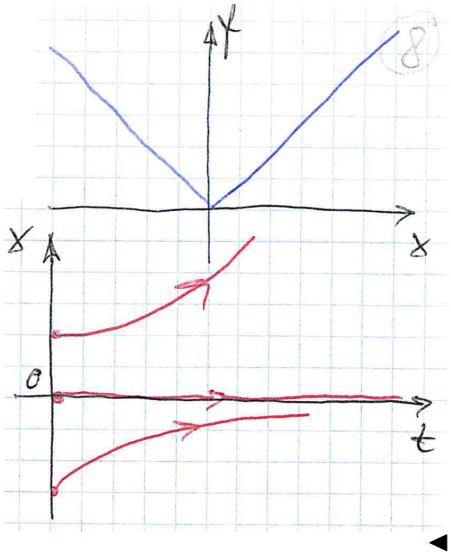
Example Unique solution

$$\dot{x} = |x|, \qquad x(t_0) = x_0$$

Solution

$$x(t) = \begin{cases} x_0 e^{t_0 - t}, & x_0 < 0, \\ 0, & x_0 = 0, \\ x_0 e^{t - t_0}, & x_0 > 0. \end{cases}$$

Note that f is continuous but not differentiable.

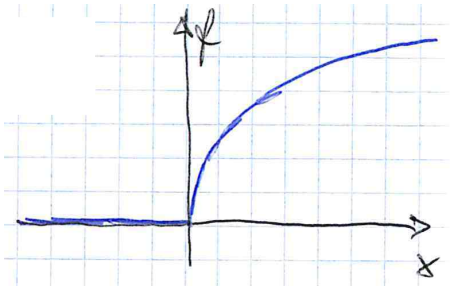


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Example Non-unique solution

$$\dot{x} = \begin{cases} \sqrt{x}, & x \geq 0, \\ 0, & x < 0, \end{cases} \qquad x(0) = 0$$

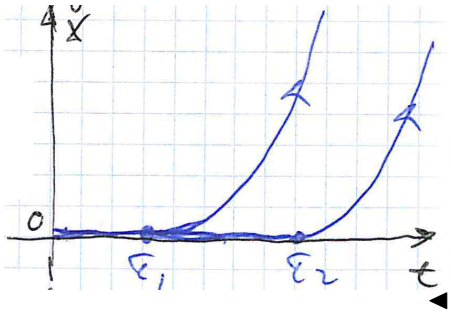
Can check that



$$x(t) = \begin{cases} 0, & t < \tau, \\ \frac{1}{4}(t - \tau)^2, & t \geq \tau \end{cases} \qquad \text{for any } \tau \geq 0$$

satisfies the equation and initial condition. But τ can be anything so a family of solutions!

Here again f is continuous and not differentiable.



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Why is the difference? Because $\dfrac{d}{dx}(\sqrt{x})$ is not bounded at 0!

To guarantee uniqueness, need f to be not just continuous, but also Lipschitz.

Definition Lipschitz property

A function $\bar{f}$ defined on a subset of $\mathbb{R}^n$ satisfies Lipschitz condition at a point $\bar{x}_0$ with Lipschitz constant L if $\exists\,(L, a)$ such that $\forall\,\bar{x}, \bar{y}$: $|\bar{x} - \bar{x}_0| < a$, $|\bar{y} - \bar{x}_0| < a$ we have

$$|\bar{f}(\bar{x}) - \bar{f}(\bar{y})| \leq L|\bar{x} - \bar{y}|.$$

Note that Differentiable $\rightarrow$ Lipschitz $\rightarrow$ Continuous

Theorem Uniqueness/Picard–Lindelöf/Cauchy–Lipschitz

Consider an IVP $\dot{\bar{x}} = \bar{f}(\bar{x}, t)$ with $\bar{x} = \bar{x}_0$ at $t = t_0$. If $\bar{f}$ satisfies a Lipschitz condition at $\bar{x}_0$ then the solution $\phi_{t-t_0}(\bar{x}_0)$ exists and is unique and continuous in a neighborhood of $(\bar{x}_0, t_0)$.

However, in general uniqueness & continuity do not necessarily mean existence of solution for all time!

Example Finite time blowup

$$\dot{x} = x^2, \qquad x(0) = 1, \qquad \text{solution is } x(t) = \frac{1}{1-t},$$

blows up to ∞ at $t \rightarrow 1$.

In what follows, unless stated otherwise, we will consider only differentiable functions f.

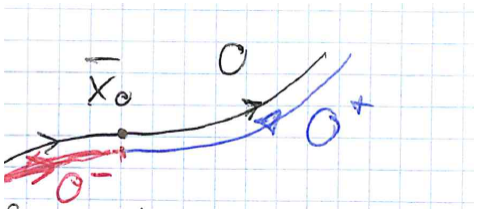
§1.3. Orbits, invariant sets and limit sets

Consider a flow $\phi_t(\bar{x}_0)$.

Definition

* The orbit/trajectory of ϕ_t through $\bar{x}_0$ is the set

$$O(\bar{x}_0) = \{\phi_t(\bar{x}_0) : -\infty < t < \infty\}.$$



* The forward orbit of ϕ_t through $\bar{x}_0$ is the set

$$O^+(\bar{x}_0) = \{\phi_t(\bar{x}_0) : 0 \le t\};$$

analogously, backward orbit $O^-(\bar{x}_0)$ if for $t \le 0$.

Definition

A set of points $\mathcal{M} \subseteq E$ is invariant under f if

$$\bar{x} \in \mathcal{M} \quad \Rightarrow \quad O(\bar{x}) \subseteq \mathcal{M}.$$

Clearly, any orbit is invariant.

Definition Fixed point

The point $\bar{x}_0$ is a fixed point (equilibrium, stationary, critical point) if $f(\bar{x}_0) = 0$. Then $\bar{x} = \bar{x}_0$ for any t and $O(\bar{x}_0) = \bar{x}_0$.

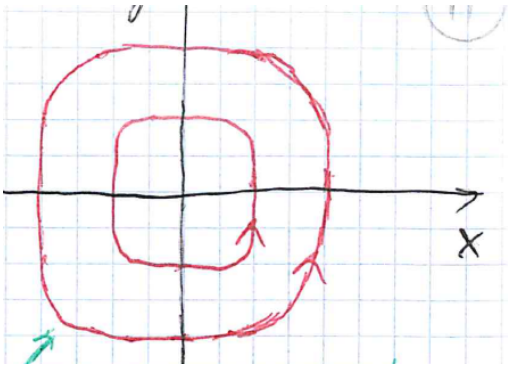
Definition Periodic point & orbit

A point $\bar{x}_0$ is a periodic point with period T if $\phi_T(\bar{x}_0) = \bar{x}_0$ for some $T > 0$ but $\phi_t(\bar{x}_0) \neq \bar{x}_0$ for $0 < t < T$.

The set $\{\phi_t(\bar{x}_0) : 0 \le t < T\}$ is the periodic orbit through $\bar{x}_0$.

Example Periodic orbit

$$\begin{pmatrix}\dot{x}\\ \dot{y}\end{pmatrix} = \begin{pmatrix}-y^3\\ x^3\end{pmatrix} \quad \Rightarrow \quad \underline{x^4 + y^4 = \text{cst}(t)}$$



not a limit cycle!

Definition Limit cycle

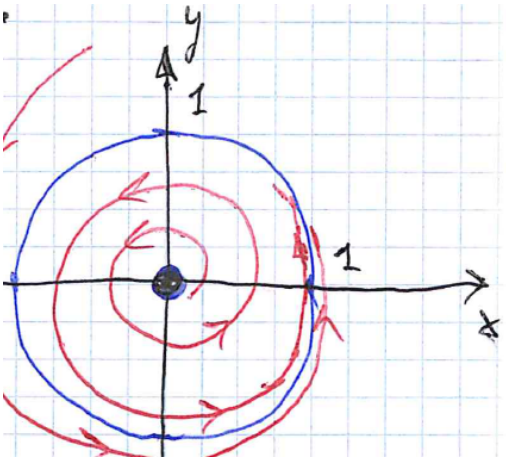
A limit cycle is an isolated periodic orbit, i.e. there are no other periodic orbits in its sufficiently small neighborhood.

Example Limit cycle

$$\begin{pmatrix}\dot{x}\\ \dot{y}\end{pmatrix} = \begin{pmatrix}-y + x(1-x^2-y^2)\\ x + y(1-x^2-y^2)\end{pmatrix} \quad \Rightarrow$$

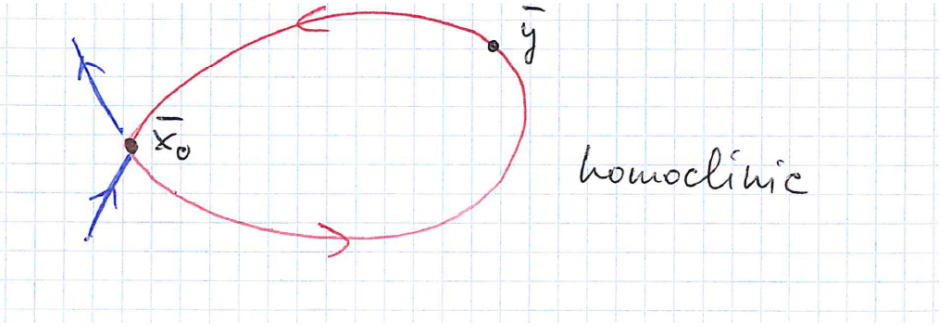
$$\underline{\dot{r} = r(1-r^2)}, \qquad \underline{\dot{\theta} = 1}.$$

A fixed point $(0,0)$ and a limit cycle $r = 1$.



Definition Homoclinic orbit

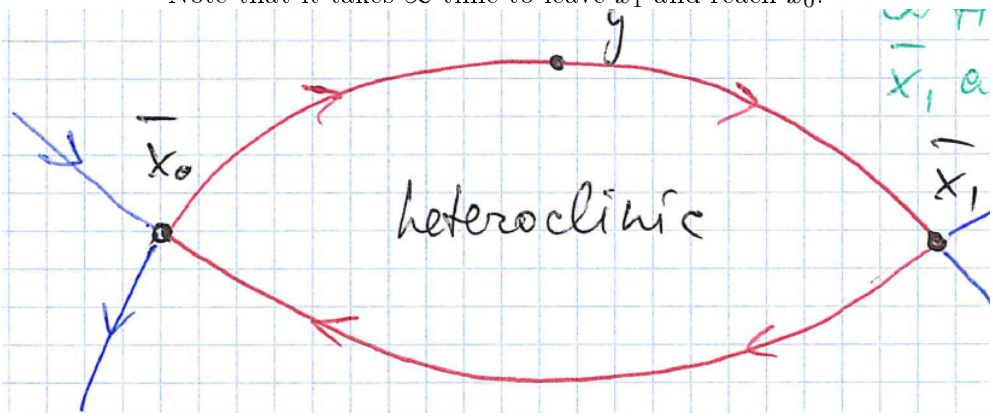
If $\bar{x}_0$ is a fixed point and $\exists\, \bar{y} \neq \bar{x}_0$ such that $\phi_t(\bar{y}) \to \bar{x}_0$ as $t \to \infty$ and $t \to -\infty$, then $O(\bar{y})$ is a homoclinic orbit.



Definition Heteroclinic orbit

If $\bar{x}_0$ and $\bar{x}_1 \neq \bar{x}_0$ are fixed points and $\exists\, \bar{y}$ such that $\phi_t(\bar{y}) \to \bar{x}_0$ as $t \to \infty$ and $\phi_t(\bar{y}) \to \bar{x}_1$ at $t \to -\infty$, then $O(\bar{y})$ is a heteroclinic orbit.

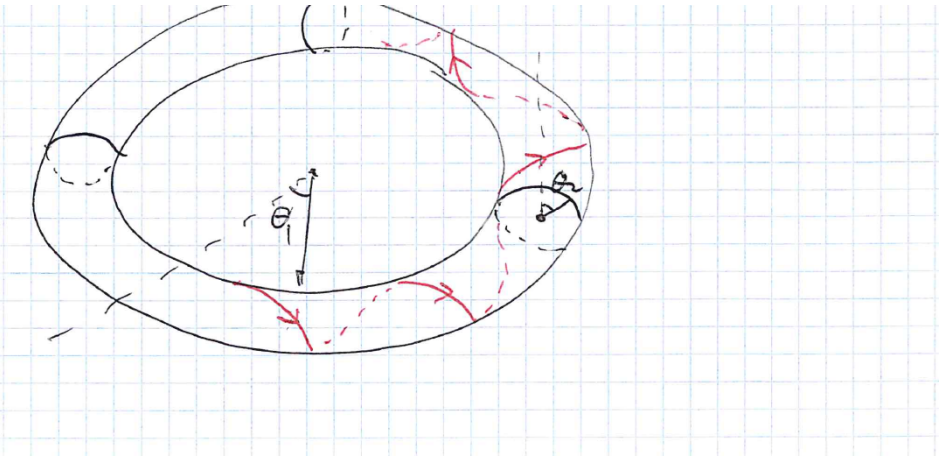
Note that it takes ∞ time to leave $\bar{x}_1$ and reach $\bar{x}_0$!



In higher dimensions more complex invariant sets are possible.

Example 2-torus in 3D

θ_1, θ_2 coordinates on the surface of a torus in 3D, $\dot{\theta}_1 = \omega_1$, $\dot{\theta}_2 = \omega_2$ and ω_1, ω_2 are not in a ratio, $\Rightarrow$ trajectory will fill the full surface of a torus. Thus, surface is an invariant set.



When talking about the long-term behavior of trajectories need to be careful with statements like $\phi_t(\bar{x}) \to \bar{y}$ as such limit may not exist, e.g. for periodic orbits.

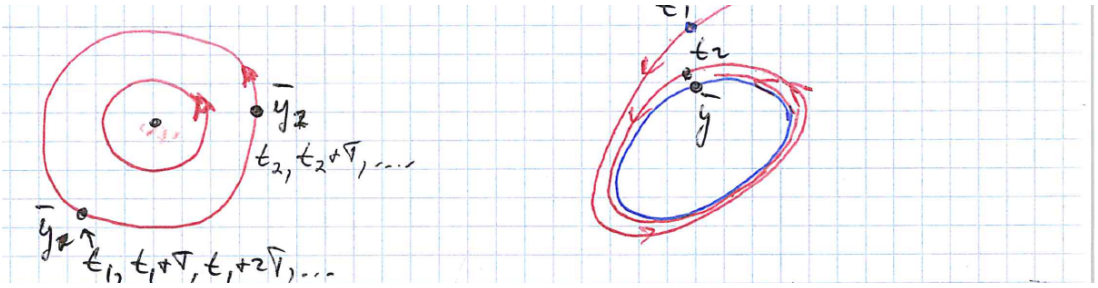
Instead use

Definition

ω-limit set of $\bar{x}$ is

$$\omega(\bar{x}) = \{\bar{y} : \exists \text{ some sequence } t_1, t_2, \ldots \to \infty \text{ with } \phi_{t_n}(\bar{x}) \to \bar{y}\}\,.$$

α-limit set defined similarly but for $t \to -\infty$.



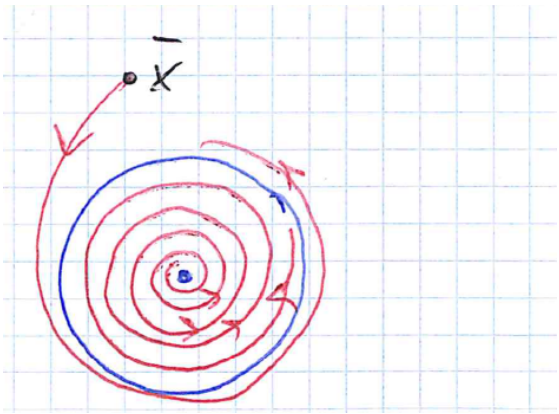
When the orbit $O(\bar{x})$ is bounded, $\omega(\bar{x})$ has properties:

* non-empty
* closed
* connected
* invariant under f
* bounded

Example

Go back to

$$\dot{r} = r(1 - r^2), \qquad \dot{\theta} = 1.$$



Here $\omega(\bar{x}) = \{r = 1\}$,

$\alpha(\bar{x}) = \{(0,0)\}$ for $|\bar{x}| < 1$ or undefined for $|\bar{x}| > 1$.

§1.4. Topological equivalence of flows

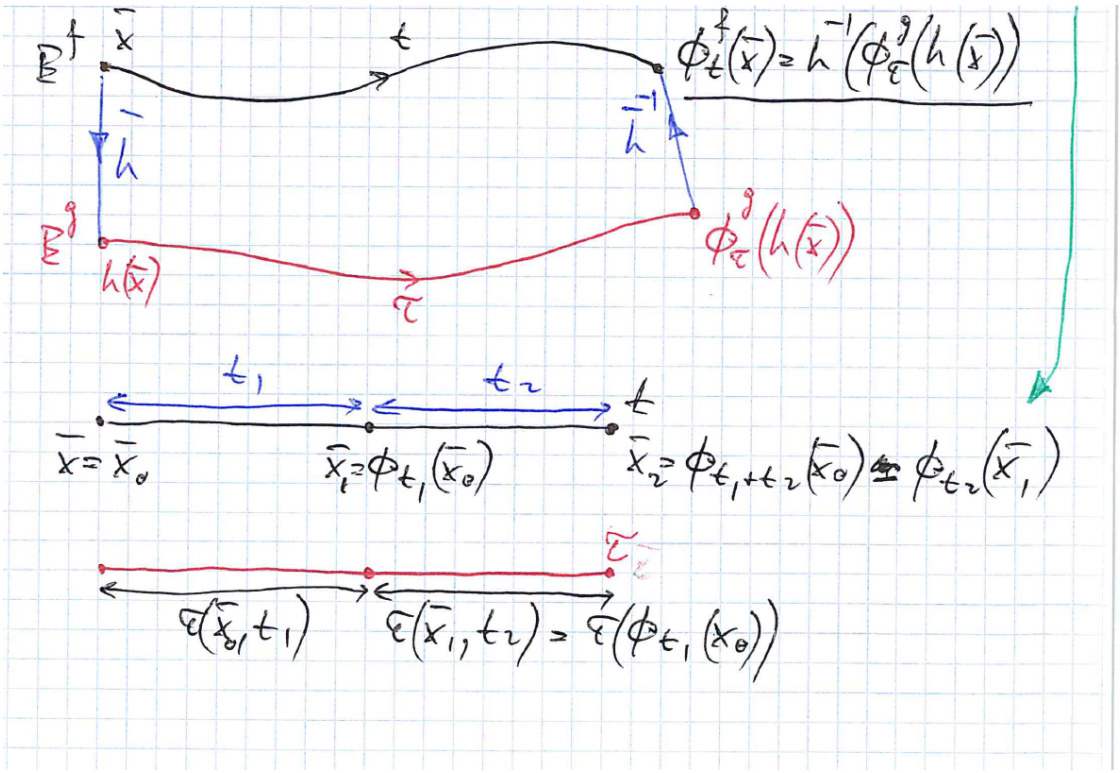
In this course we will sometimes be referring to different dynamical systems/flows as topologically/geometrically equivalent. Later we'll see that the flow structure changes at bifurcation.

Definition Topological equivalence

Two flows $\phi_t^f(\bar{x})$ and $\phi_t^g(\bar{x})$ are topologically equivalent if there is a homeomorphism $h(\bar{x}) : E^f \to E^g$ (i.e. a continuous bijection with continuous inverse) and time-increasing function $\tau(\bar{x}, t)$ such that for any $t, \bar{x}$

$$\phi_t^f(\bar{x}) = h^{-1} \circ \phi_\tau^g \circ h(\bar{x})$$

$$\text{and} \qquad \tau(\bar{x}, t_1 + t_2) = \tau(\bar{x}, t_1) + \tau\big(\phi_{t_1}^f(\bar{x}), t_2\big).$$



End of lecture 3

§1.4. Topological equivalence of flows

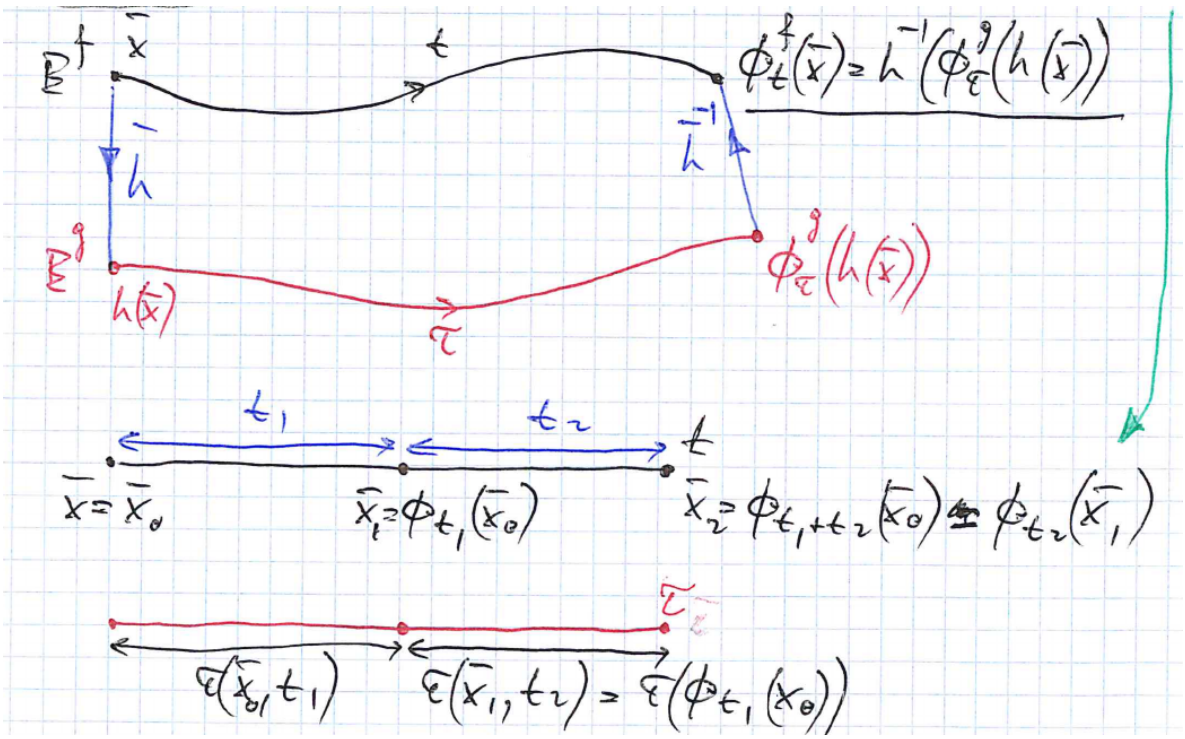
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$$\phi_t^f(\bar{x}) = \bar{h}^{-1} \circ \phi_\tau^g \circ h(\bar{x})$$

and $$\tau(\bar{x}, t_1 + t_2) = \tau(\bar{x}, t_1) + \tau\big(\phi_{t_1}^f(\bar{x}), t\big).$$



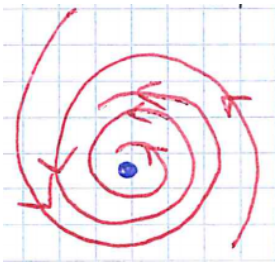
In words, $\exists$ a 'nice' mapping from one phase-space to the other and a map from one time to another such that the evolutions of the two systems are the same.

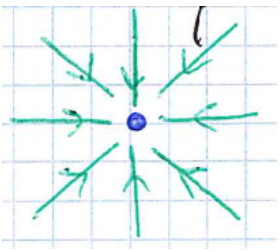
Clearly, $\bar{h}$ must map fixed points into fixed points and periodic orbits to periodic orbits (with generally different period).

If $\tau = t$ we say that the two flows are topologically conjugate.

Example

$$\begin{aligned} \dot{r} &= -r \\ \dot{\theta} &= 1 \end{aligned} \qquad \text{and} \qquad \begin{aligned} \dot{\rho} &= -2\rho \\ \dot{\psi} &= 0 \end{aligned}$$





Flows are topologically equivalent with $\bar{h}(0,0) = (0,0)$, $h(r,\theta) = (r^2, \theta + \ln r)$ for $r \neq 0$ in polars and $\tau = t$ (so topologically conjugate).

To show, solve both: starting at $\bar{x} = (r_0, \theta_0)$

$$\phi_t^f = (r_0 e^{-t}, \theta_0 + t), \qquad \phi_t^g = (\rho_0 e^{-2t}, \psi_0).$$

Then

$$h \circ \phi_t^f = \left(r_0^2 e^{-2t}, \theta_0 + t + \ln(r_0 e^{-t})\right) = \left(r_0^2 e^{-2t}, \theta_0 + \ln r_0\right)$$

and

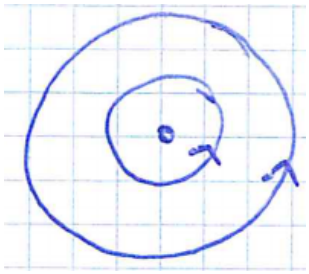
$$\phi_t^g \circ h = \left(r_0^2 e^{-2t}, \theta_0 + \ln r_0\right) \quad \text{– the same, } \Rightarrow \text{ Ok}$$

$$\text{as } h(\bar{x}) = (r_0^2, \theta_0 + \ln r_0).$$

◀

Example

$$\begin{aligned}\dot{r} &= 0 \\ \dot{\theta} &= 1\end{aligned} \qquad \text{and} \qquad \begin{aligned}\dot{r} &= 0 \\ \dot{\theta} &= r + \sin^2\theta\end{aligned}$$



Both are periodic circular orbits, so $\bar{h}(\bar{x}) = \bar{x}$, and time is stretched:

$$\frac{d\theta}{dt} * (r + \sin^2\theta) = \frac{d\theta}{d\tau}, \quad \Rightarrow$$

$$\tau = \int_0^t \frac{dt'}{r_0 + \sin^2(\theta_0 + t')}$$

as first flow is

$$\phi_t(r_0, \theta_0) = (r_0, \theta_0 + t).$$

Thus, topologically equivalent (but not conjugate). ◀

Definition

The vector field $\bar{f}$ (or system $\dot{\bar{x}} = \bar{f}(\bar{x})$ or flow $\phi_t(\bar{x})$) is structurally stable if $\exists\, \varepsilon > 0$ such that $\bar{f} + \bar{\delta}$ is topologically equivalent to $f\ \forall\, \bar{\delta}(\bar{x})$ with

$$|\bar{\delta}| + \sum_i \left| \frac{\partial \bar{\delta}}{\partial x_i} \right| < \varepsilon.$$

In the first example system is structurally stable, in the second it is not (a small $\delta \neq 0$ will turn $\dot{r} = 0$ into $\dot{r} = \delta$, which is no longer a periodic orbit).

We will discuss this later in the context of bifurcations.

§2. Behavior near fixed points

Suppose $\bar{x}_0$ is a fixed point of $\dot{\bar{x}} = \bar{f}(\bar{x})$, i.e. $\bar{f}(\bar{x}_0) = 0$. If f is smooth near $\bar{x}_0$, set

$$\bar{y} = \bar{x} - \bar{x}_0, \quad \Rightarrow \quad \dot{\bar{y}} = (\bar{x} - \dot{\bar{x}}_0) = \dot{\bar{x}} = \bar{f}(\bar{x}) = \bar{f}(\bar{y} + \bar{x}_0).$$

Taylor expand r.h.s.:

$$\dot{\bar{y}} = \underbrace{\bar{f}(\bar{x}_0)}_{=0} + \hat{A}\bar{y} + O(|y|^2), \qquad A_{ij} = \frac{\partial f_i}{\partial x_j} \quad \begin{matrix} \text{– the Jacobian} \\ \text{matrix of } f \text{ at } \bar{x}_0. \end{matrix}$$

Thus, near $\bar{x}_0$ the behavior of the system is approximated by $\dot{\bar{y}} = \hat{A}\bar{y}$.

§2.1 2D flows near fixed points

In 2D is a 2×2 matrix $\hat{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $\mathrm{Det}(\hat{A}) = ad - bc$, $\mathrm{Tr}(\hat{A}) = a + d$.

Eigenvalues satisfy characteristic eq-n

$$\lambda^2 - \lambda \cdot \mathrm{Tr}(\hat{A}) + \det(\hat{A}) = 0, \quad \Rightarrow \quad \lambda = \frac{1}{2}\left[\mathrm{Tr}(\hat{A}) \pm \sqrt{(\mathrm{Tr}(\hat{A}))^2 - 4\,\mathrm{Det}(\hat{A})}\right].$$

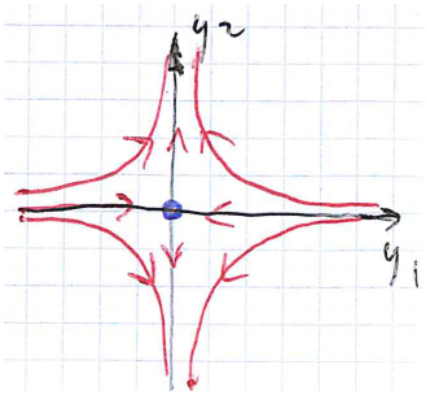
By finding eigenvectors (when we can) we can put $\hat{A}$ into canonical form & analyze.

① Saddle point: $\mathrm{Det}(\hat{A}) < 0$, $\Rightarrow \lambda_1, \lambda_2$ are real and have opposite signs, $\lambda_1\lambda_2 < 0$.

$$\hat{A} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}, \quad \text{e.g. } \lambda_1 < 0 < \lambda_2$$

$$\Updownarrow$$

$$\dot{y}_1 = \lambda_1 y_1, \quad \dot{y}_2 = \lambda_2 y_2, \quad \Rightarrow \quad \begin{matrix} y_1 = y_{1,0} e^{\lambda_1 t} \\ y_2 = y_{2,0} e^{\lambda_2 t} \end{matrix}$$



End of lecture 4

Since, eliminating t, $y_2 \propto y_1^{\lambda_2/\lambda_1} = \dfrac{1}{y_1^{|\lambda_2/\lambda_1|}}$. So, orbits are hyperbolas!

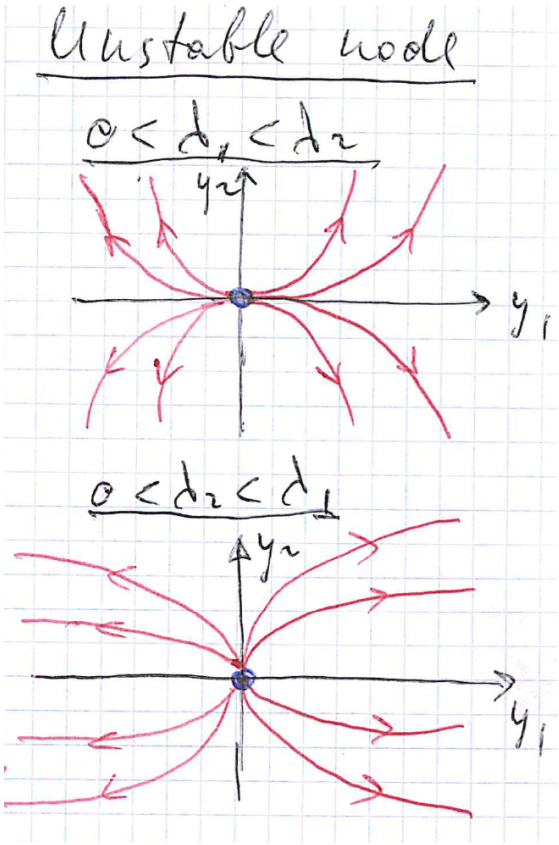
② Node: $(\mathrm{Tr}(\hat{A}))^2 > 4\,\mathrm{Det}(\hat{A}) > 0$, $\Rightarrow \lambda_1, \lambda_2$ are real and either both positive ($\mathrm{Tr}(\hat{A}) > 0$) or negative ($\mathrm{Tr}(\hat{A}) < 0$).

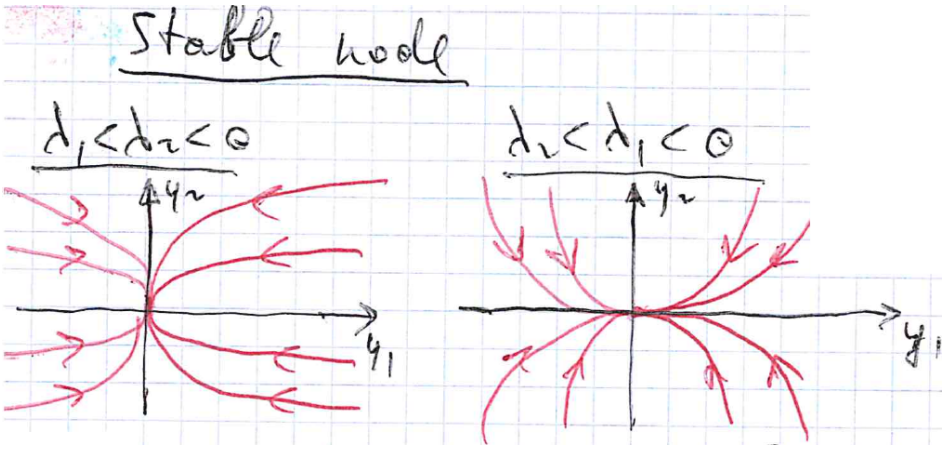
Same canonical form

$$\hat{A} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}, \quad \Rightarrow \quad y_2 \propto y_1^{\lambda_2/\lambda_1}.$$

Get different patterns depending on $\lambda_2/\lambda_1 < 1$ or > 1.

Sign of λ_1, λ_2 sets the direction of the flow.





③ Focus (spiral): $(\mathrm{Tr}(\hat{A}))^2 < 4\,\mathrm{Det}(\hat{A})$ and $\det(\hat{A}) > 0$ so that roots are complex, $\lambda = \rho \pm i\omega$, $\omega \neq 0$.

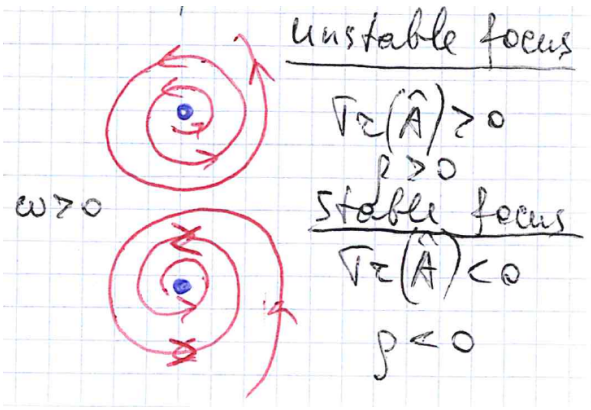
Canonical form

$$\hat{A} = \begin{pmatrix} \rho & -\omega \\ \omega & \rho \end{pmatrix}.$$

In polars have

$$\dot{r} = \rho r, \quad \dot{\theta} = \omega, \quad \Rightarrow$$

$$r = r_0 e^{\rho t}, \quad \theta = \theta_0 + \omega t.$$



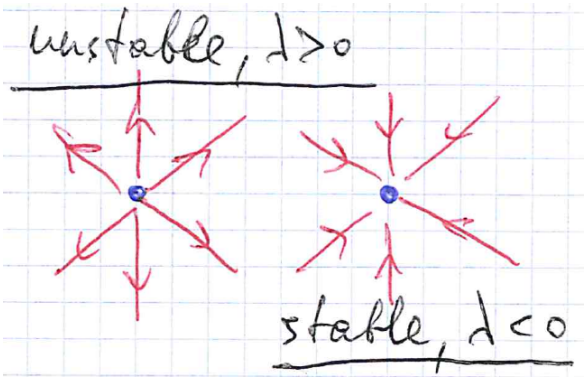
Other cases are degenerate.

④ $(\mathrm{Tr}(\hat{A}))^2 = 4\,\mathrm{Det}(\hat{A}) > 0$

A boundary between focus & node, double eigenvalue $\lambda = \frac{1}{2}\mathrm{Tr}(\hat{A})$. Two cases:

Stellar node (star)

if $\hat{A} = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$ is canonical



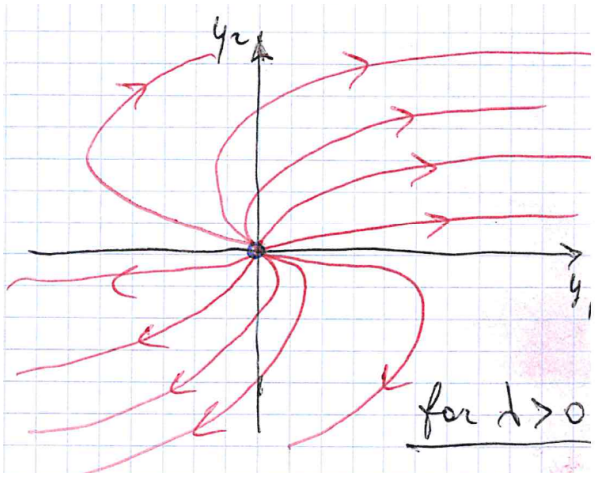
Improper node

if $\hat{A} = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$ is canonical

(defective)

Here $y = y_0 e^{\lambda t}$,

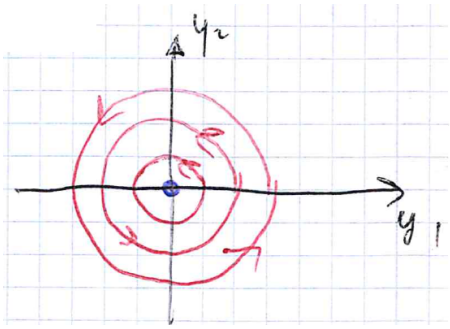
$x = x_0 e^{\lambda t} + y_0 t e^{\lambda t}$, $\quad \dfrac{y}{x} = \dfrac{y_0}{x_0 + y_0 t}$.



⑤ $\mathrm{Det}(\hat{A}) > 0$, $\quad \mathrm{Tr}(\hat{A}) = 0$: Centre

Here $\lambda = \pm i\omega$, $\qquad \hat{A} = \begin{pmatrix} 0 & -\omega \\ \omega & 0 \end{pmatrix}$.

$\dot{r} = 0$, $\quad \dot{\theta} = \omega$, $\quad \Rightarrow$ orbits are closed circles.

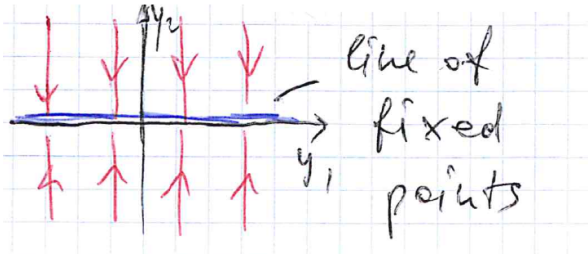


⑥ $\mathrm{Det}(\hat{A}) = 0$, $\quad \mathrm{Tr}(\hat{A}) \neq 0$ – between saddle & node

$\lambda_1 = 0$, $\quad \lambda_2 \neq 0$, $\qquad \hat{A} = \begin{pmatrix} 0 & 0 \\ 0 & \lambda \end{pmatrix}$

$y_1 = \text{const}$, $\qquad y_2 = y_{2,0} e^{\lambda t}$

for $\lambda < 0$

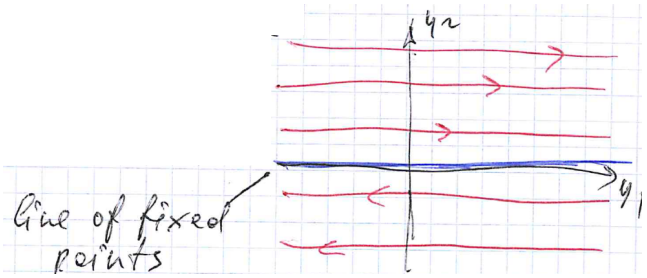


⑦ $\underline{\mathrm{Det}(\hat{A}) = 0, \quad \mathrm{Tr}(\hat{A}) = 0}$, $\lambda_{1,2} = 0$

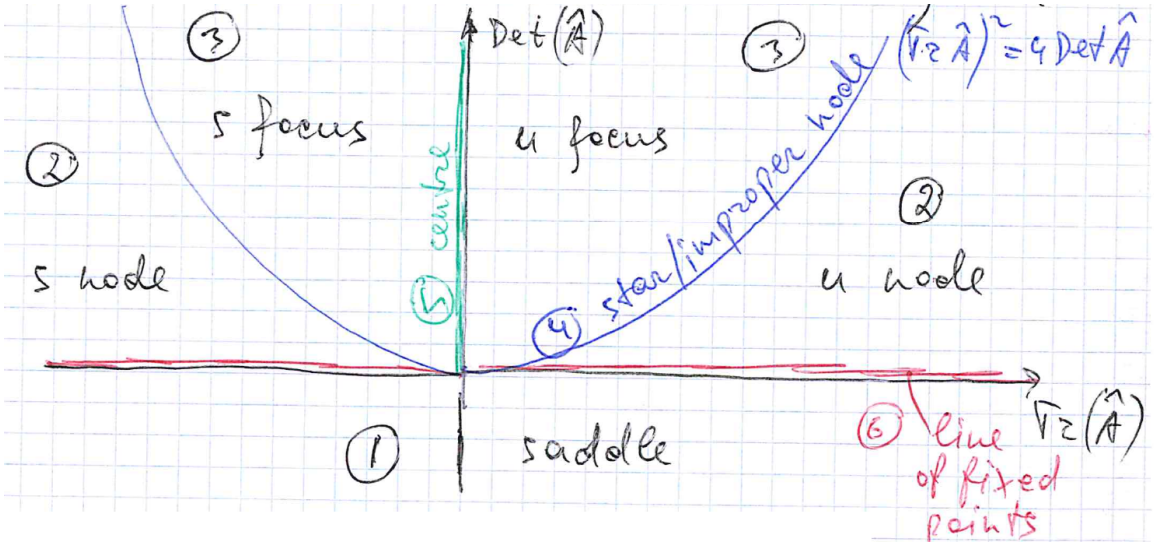
if $\hat{A} = 0$, $\Rightarrow y_1 = \mathrm{const}$, $y_2 = \mathrm{const}$, $\Rightarrow$ plane of fixed points!

if $\hat{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, this is a case of shear,

$$\underline{\dot{y}_1 = y_2, \quad \dot{y}_2 = 0.}$$



This zoo can be summarized in a diagram



Canonical forms are in special basis

$$(*) \quad \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \quad \text{– eigenvectors}$$

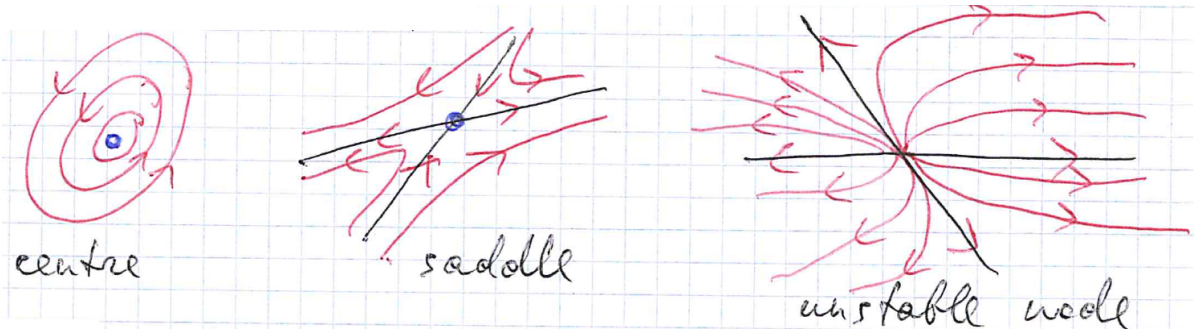
$$(*) \quad \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} \quad \text{– generalized eigenvectors}$$

$$(*) \quad \begin{pmatrix} \rho & -\omega \\ \omega & \rho \end{pmatrix} \quad \text{– Re \& Im of eigenvectors}$$

Notes

(∗) No need to change basis in practice, just work with $\mathrm{Tr}(\hat{A})$, $\mathrm{Det}(\hat{A})$ to classify

(∗) In original (non-canonical) coordinates shapes can be distorted



(∗) Useful to know eigenvectors for saddles/nodes; for focus/centre cases ratios depend also on $\lambda_{1,2}$, easier to just plot several values

(∗) Centres are rare but generic in Hamiltonian systems:

$$\dot{\bar{x}} = \begin{pmatrix} \dfrac{\partial H}{\partial y} \\ -\dfrac{\partial H}{\partial x} \end{pmatrix} \quad \text{for some } H(x,y), \quad \bar{x} = (x,y).$$

$$\text{Indeed,} \quad \hat{A} = \begin{pmatrix} H_{xy} & H_{yy} \\ -H_{xx} & -H_{xy} \end{pmatrix}, \quad \Rightarrow \quad \underline{\mathrm{Tr}(\hat{A}) = 0}, \quad \Rightarrow$$

$$\text{either} \quad \underset{\lambda = \pm i\omega}{\text{centres}} \quad \text{or} \quad \underset{\lambda_1 \lambda_2 < 0}{\text{saddles}}.$$

$$\text{Also,} \quad \underline{\frac{dH}{dt}} = (\dot{\bar{x}} \cdot \nabla) H = \dot{x} H_x + \dot{y} H_y = H_y H_x - H_x H_y = 0.$$

Thus, $H(x,y) = \text{const}$ on trajectories, orbits follow contours of $H(x,y)$ in autonomous Hamiltonian systems.

End of lecture 5

§2.2 Effect of nonlinear terms

How does the behavior of linearization $\dot{\bar{y}} = \hat{A}\bar{y}$ of the system $\dot{\bar{x}} = \bar{f}(\bar{x})$ about the fixed point $\bar{x}_0$ inform us about the behavior of the full nonlinear system around $\bar{x}_0$?

Definition

A fixed point $\bar{x}_0$ is hyperbolic if none of the eigenvalues of the Jacobian of $\bar{f}(\bar{x})$ (matrix $\hat{A}$) at $\bar{x}_0$ have zero real part. It is non-hyperbolic otherwise.

We classify hyperbolic fixed points as

- $(*)$ source if all $\mathrm{Re}(\lambda_i) > 0$
- $(*)$ sink if all $\mathrm{Re}(\lambda_i) < 0$
- $(*)$ saddle otherwise (i.e. some $\mathrm{Re}(\lambda_i) > 0$, others < 0).

We will discuss separately hyperbolic & non-hyperbolic fixed points.

One can show (e.g. see Glendinning, Ch. 4) that if

(1) $\bar{x}_0$ is hyperbolic and

(2) the nonlinear corrections of $\bar{f}$ are $O(|\bar{x} - \bar{x}_0|^2)$

then the full nonlinear system and the linearized system are topologically conjugate.

Note: non-hyperbolic fixed points arise when

$$\{\mathrm{Det}\,\hat{A} \geq 0,\ \mathrm{Tr}\,\hat{A} = 0\}, \quad \{\mathrm{Det}\,\hat{A} = 0,\ \mathrm{Tr}\,\hat{A} \neq 0\},$$

i.e. cases 5, 6, 7 identified earlier.

§2.2.1 Stable and unstable manifolds

For a linearized system we can separate the phase space into different domains according to different behaviors in time.

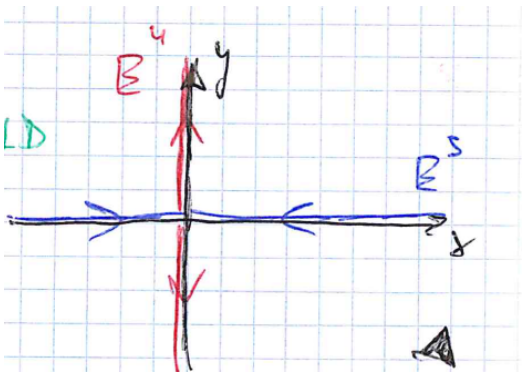
Definition

The stable, unstable and centre subspaces of the linearization of $\bar{f}(\bar{x})$ near a fixed point $\bar{x}_0$ are the three linear subspaces E^s, E^u and E^c spanned by subsets of (possibly generalized) eigenvectors of $\hat{A}$ whose eigenvalues have real parts $\mathrm{Re}(\lambda) < 0$, $\mathrm{Re}(\lambda) > 0$ and $\mathrm{Re}(\lambda) = 0$, respectively.

Note: a hyperbolic fixed point has no centre subspace.

Example

$$\underset{\text{2D}}{\hat{A} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}}, \quad \Rightarrow \quad \begin{matrix} E^s = x\text{-axis} \\ E^u = y\text{-axis} \end{matrix} \quad \text{1D}$$



Theorem Stable manifold theorem

Suppose $\bar{0}$ is a hyperbolic fixed point of $\dot{\bar{x}} = \bar{f}(\bar{x})$ and E^s and E^u are the stable and unstable subspaces of the linearization of $\bar{f}$ near $\bar{0}$. Then $\exists$ stable and unstable manifolds W^s_{loc} & W^u_{loc}, which have the same dimension as E^s, E^u and are tangent to E^s & E^u at $\bar{0}$, such that for $\bar{x} \neq 0$ but in a sufficiently small neighborhood of $\bar{0}$,

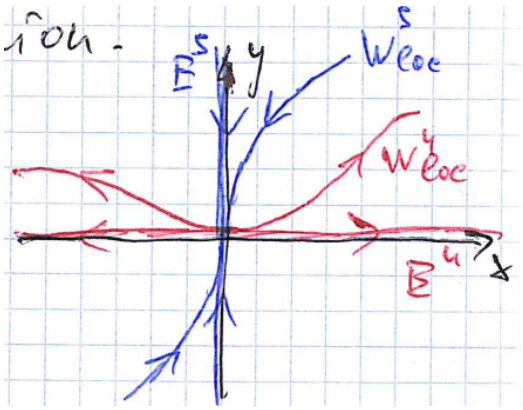
$$W^s_{\text{loc}} = \{\bar{x} : \phi_t(\bar{x}) \to \bar{0} \text{ as } t \to +\infty\},$$

$$W^u_{\text{loc}} = \{\bar{x} : \phi_t(\bar{x}) \to \bar{0} \text{ as } t \to -\infty\}.$$

Proof: not part of the course, see Glendinning.

The local stable manifold W^s_{loc} can be extended to a global invariant manifold W^s by following the flow backwards in time starting at $\bar{x} \in W^s_{\text{loc}}$. Same for W^u_{loc}, extend to W^u by $t \to +\infty$.

In $\mathbb{R}^2$, this theorem guarantees the existence of two trajectories that approach and leave a saddle fixed point in a tangent fashion.



It is often easy in practice to find local approximations to stable and unstable manifolds.

Indeed, for a saddle in $\mathbb{R}^2$ assume that E^s is $x = 0$ and E^u is $y = 0$ (can always change the basis).

Then look for W^s_{loc} in the form $x = S(y)$ with $S(0) = 0$ and $S'(0) = 0$. For W^u_{loc} take $y = U(x)$ with $U(0) = 0$, $U'(0) = 0$. These manifolds must satisfy $\dot{\bar{x}} = \bar{f}(\bar{x})$, i.e. $\dot{x} = \frac{\partial S}{\partial y}\dot{y}$ and $\dot{y} = \frac{\partial U}{\partial x}\dot{x}$ must solve $\dot{\bar{x}} = \bar{f}(\bar{x})$. Thus, for

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} f(x,y) \\ g(x,y) \end{pmatrix},$$

$$\dot{x} = \frac{dS}{dy}\dot{y} = \underbrace{\frac{dS}{dy} g(S(y),y) = f(S(y),y)}_{\text{(nonlinear) ODE for } S(y)} \quad \text{and}$$

$$\dot{y} = \frac{dU}{dx}\dot{x} = \underbrace{\frac{dU}{dx} f(x,U(x)) = g(x,U(x))}_{\text{ODE for } U(x)}.$$

No need to solve these ODEs, we can simply find a locally convergent series expansion in the form

$$S(y) = a_2 y^2 + a_3 y^3 + \cdots, \qquad U(x) = b_2 x^2 + b_3 x^3 + \cdots$$

and solve term by term.

Example

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} x(1-y) \\ -y + x^2 \end{pmatrix}, \qquad \hat{A} = \begin{pmatrix} 1-y & -x \\ 2x & -1 \end{pmatrix}\Big|_{(0,0)} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

$$\det \hat{A} = -1 \;\Rightarrow\; \underline{\text{saddle}}, \qquad \Rightarrow \quad E^s = \{x = 0\}, \quad E^u = \{y = 0\}.$$

For W^u_{loc} take

$$y = U(x) = a_2 x^2 + a_3 x^3 + a_4 x^4 + \cdots,$$

substitute into

$$\dot{y} = -y + x^2 = \frac{dU}{dx} x(1-y), \quad \Rightarrow$$

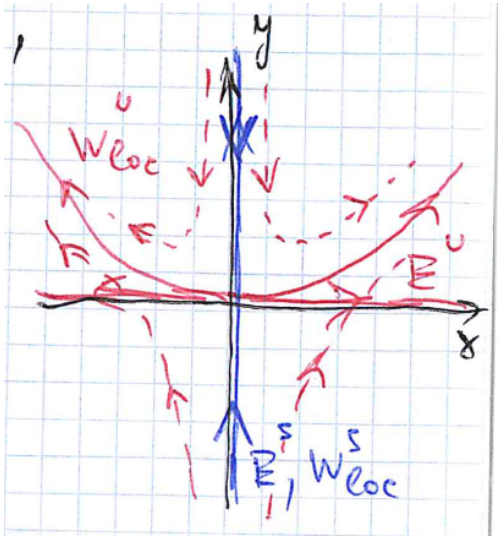
$$-U(x) + x^2 = \frac{dU}{dx} x(1 - U(x)), \quad \Rightarrow$$

$$x^2 - a_2 x^2 - a_3 x^3 - a_4 x^4 + \cdots = (2a_2 x + 3a_3 x^2 + 4a_4 x^3 + \cdots)\, x \times (1 - a_2 x^2 - a_3 x^3 - a_4 x^4 + \cdots), \quad \Rightarrow$$

$$\begin{aligned} x^2: &\quad 1 - a_2 = 2a_2 &&\Rightarrow \quad a_2 = \frac{1}{3} \\ x^3: &\quad -a_3 = 3a_3 &&\Rightarrow \quad a_3 = 0 \\ x^4: &\quad -a_4 = 4a_4 - 2a_2^2 &&\Rightarrow \quad a_4 = \frac{2}{45} \end{aligned}$$

due to $x \mapsto -x$ symmetry! $\quad (a_3 = 0)$

$$\text{Thus,} \quad U(x) = \frac{x^2}{3} + \frac{2}{45} x^4 + \cdots.$$



To find W^s_{loc}, $x = S(y) = a_2 y^2 + a_3 y^3 + \cdots$,

$$S(y)(1-y) = \frac{dS}{dy}[-y + S^2(y)],$$

repeat calculation, find $a_n = 0\ \forall n$. This is caused by symmetry! If $X = -x$ then system the same so W^s_{loc} has to be symmetric w.r.t. y-axis & tangent to it,

$$\Rightarrow \quad W^s_{\text{loc}} = y\text{-axis}.$$

End of lecture 6

Note:

- If E^s & E^u are not aligned with the coordinate axes, one can either transform the coordinates to get alignment, or consider functions S & U tangent to the relevant eigenvector. E.g. for $\bar{e} = \binom{1}{k}$ use $y = kx + a_2x^2 + a_3x^3 + \cdots$.
- In higher dimensions use multi-coordinate versions of S & U. E.g. for a 2D E^s in 3D use S: $z = S(x, y)$ and $\dot{z} = \dot{x}\dfrac{\partial S}{\partial x} + \dot{y}\dfrac{\partial S}{\partial y}$ and expand in both coordinates x, y.

§2.2.2 Non-linear terms in non-hyperbolic cases

Suppose at least one eigenvalue has $\mathrm{Re}(\lambda) = 0$. Have many possibilities.

(a) $\lambda = \pm i\omega$, linear system has a centre. In nonlinear case multiple possible outcomes, e.g.

* $\begin{aligned}\dot{x} &= -y - x^3,\\ \dot{y} &= x - y^3,\end{aligned}$ $\Rightarrow \quad \dot{r} = -\dfrac{x^4 + y^4}{r} \le 0,$

$$\dot{\theta} = 1 - \frac{xy^3 - yx^3}{r^2} \to 1 \quad \text{as } r \to 0.$$



$H = -\frac{1}{2}(x^2 + y^2)$ for linear.

* $\begin{aligned}\dot{x} &= -y + x^3,\\ \dot{y} &= x + y^3,\end{aligned}$ $\Rightarrow \quad \dot{r} = \dfrac{x^4 + y^4}{r} \ge 0,$

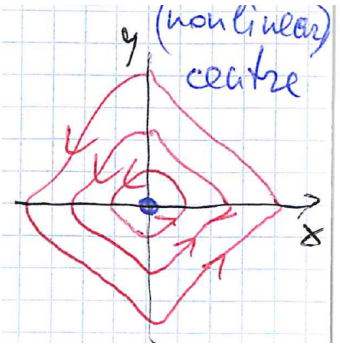
$$\dot{\theta} \to 1 \quad \text{as } r \to 0.$$



* $\begin{aligned}\dot{x} &= -y - 2x^2y,\\ \dot{y} &= x + 2y^2x\end{aligned}$ Hamiltonian system,

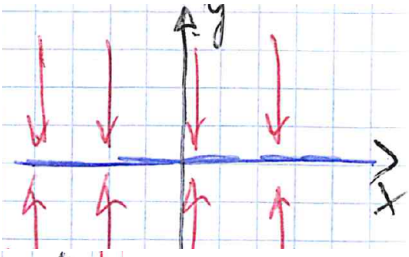
$$\dot{x} = \frac{\partial H}{\partial y}, \qquad \dot{y} = -\frac{\partial H}{\partial x},$$

with $H = -\frac{1}{2}(x^2 + y^2) - x^2y^2$ — contours are orbits.



(b) $\lambda_1 = 0$, $\lambda_2 \neq 0$, linearized — line of fixed points.

Some possibilities:



* $\begin{aligned} \dot{x} &= x^2, \\ \dot{y} &= -y, \end{aligned}$ linearized $\begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}$,

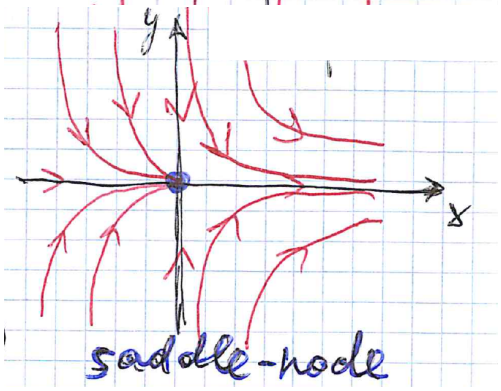
$$\text{Det}\,\hat{A} = 0, \quad \text{Tr}\,\hat{A} = -1.$$

Full solution:

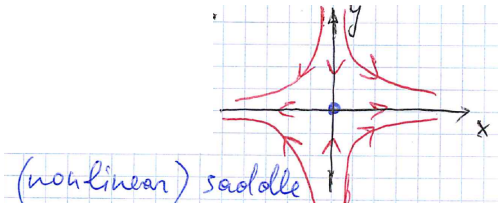
$$y \propto \exp\left(\frac{1}{x}\right), \quad \dot{x} > 0, \quad y \propto e^{-t}.$$

Note: $E^c = \{y = 0\}$.

In the nonlinear case have multiple (non-unique) manifolds tangent to E^c at $(0,0)$ — no analogue of stable manifold theorem for centre manifolds.

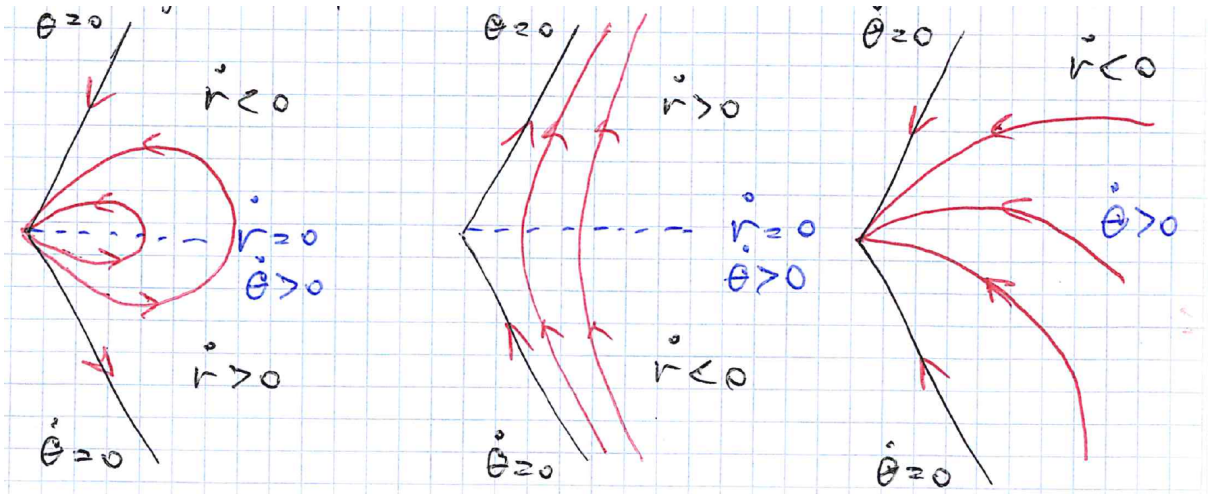


* $\begin{aligned} \dot{x} &= x^3, \\ \dot{y} &= -y, \end{aligned}$ $y \propto \exp\left(\frac{1}{2x^2}\right)$.



(c) $\lambda_1 = \lambda_2 = 0$, linearized — plane or line of fixed points.

Almost anything is possible! Change to polars, find rays $\dot{\theta} = 0$, between them can have



§2.3 Sketching phase portraits

Example $\quad \dot{x} = x(1-y), \quad \dot{y} = -y + x^2, \quad x \mapsto -x$ symmetry, $\Rightarrow$ only $x \geq 0$ is enough.

Fixed points $(0,0)$, $(1,1)$, $(-1,1)$. FP at $(0,0)$ is a saddle, studied before. Focus on $(1,1)$.

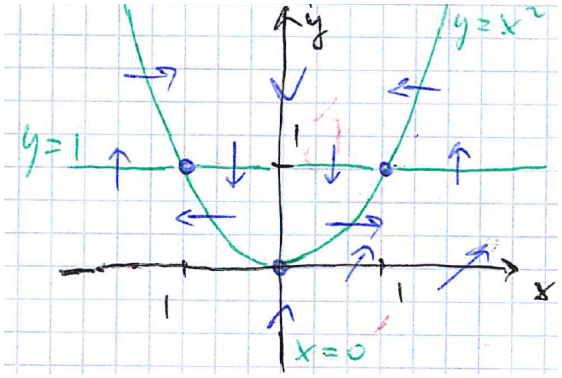
Linearize:

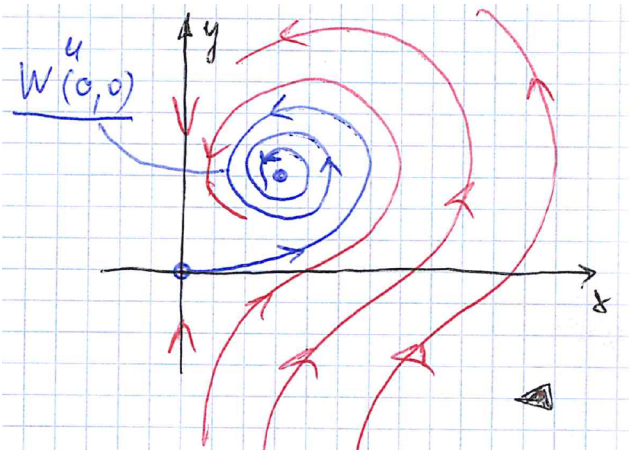
$$\hat{A} = \begin{pmatrix} 1-y & -x \\ 2x & -1 \end{pmatrix}, \qquad \hat{A}_{(1,1)} = \begin{pmatrix} 0 & -1 \\ 2 & -1 \end{pmatrix},$$

$$\operatorname{Tr}\hat{A} = -1, \quad \operatorname{Det}\hat{A} = 2, \quad (\operatorname{Tr}\hat{A})^2 < 4\operatorname{Det}\hat{A} \quad \Rightarrow \quad \text{focus},$$

$$\lambda = \rho \pm i\omega \quad \text{and} \quad \rho = \tfrac{1}{2}\operatorname{Tr}\hat{A} < 0 \quad \Rightarrow \quad \text{stable focus}.$$

Nullclines (where $\dot{x} = 0$, $\dot{y} = 0$ separately) get signs of $\dot{y}$, $\dot{x}$ on them.





General strategy

(1) Check for symmetries, find fixed points

(2) Calculate Jacobian, classify fixed points

(3) For saddles, find eigenvectors of Jacobian

(4) Find nullclines (where $\dot{x}$ or $\dot{y} = 0$) and determine flow direction on them and around

(5) Use knowledge of stable/unstable manifolds

(6) Combine these features into a global picture

(7) Trajectories cannot cross!

(8) Search for periodic orbits (see Chapter 4)

End of lecture 7

§3. Stability

If we found a fixed point, or more generally an invariant set Λ, we know that starting at Λ system will remain there. But what happens if we slightly perturb the system away from Λ? Does the system remain near Λ? Or does it converge back to or away from Λ? These are the issues of system's stability, for which there are several definitions.

§3.1 Definitions

Let Λ be an invariant set (a fixed point, periodic orbit, etc.) of a general (autonomous) system described by a flow ϕ_t.

Definition Neighborhood of a set Λ

For $\delta > 0$ the neighborhood of Λ is defined as

$$N_\delta(\Lambda) = \{\bar{x} : \exists\, \bar{y} \in \Lambda \text{ such that } |\bar{x} - \bar{y}| < \delta\}.$$

Definition Flow tending to Λ

The flow $\phi_t \to \Lambda$ if

$$\inf_{\bar{y}\in\Lambda} |\phi_t(\bar{x}) - \bar{y}| \to 0 \quad \text{as } t \to \infty.$$

Definition Lyapunov stability

The set Λ is Lyapunov stable if $\forall\, \varepsilon > 0\ \exists\, \delta > 0$ such that if $\bar{x} \in N_\delta(\Lambda)$ then $\phi_t(\bar{x}) \in N_\varepsilon(\Lambda) \quad \forall\, t \geq 0$.

"Start near, stay near"

Definition Quasi-asymptotic stability (QAS)

The set Λ is quasi-asymptotically stable if $\exists\, \delta > 0$ such that if $\bar{x} \in N_\delta(\Lambda)$ then $\phi_t(\bar{x}) \to \Lambda$ as $t \to \infty$.

"Get close eventually"

Definition Asymptotic stability

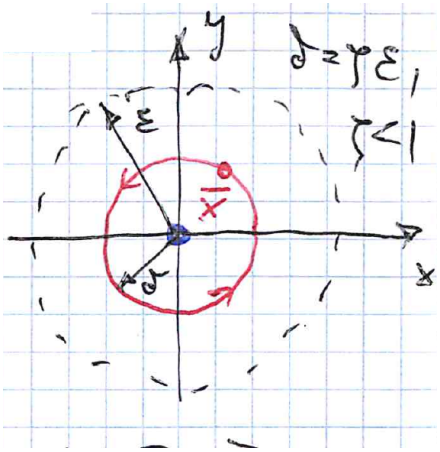
The set Λ is asymptotically stable if it is both Lyapunov stable and quasi-asymptotically stable.

"Stays near and converges to Λ"

Example Λ is LS but not QAS

$$\begin{pmatrix}\dot{x}\\ \dot{y}\end{pmatrix} = \begin{pmatrix}-y\\ x\end{pmatrix}$$

Hamiltonian system with circular orbits. For $\forall \varepsilon$ take $\delta = \zeta\varepsilon$, $0 < \zeta < 1$, then for $|\bar{x}| < \delta$ always have $|\phi_t(\bar{x})| < \varepsilon$, $\Rightarrow$ origin is LS. But no convergence to $\bar{x} = \bar{0}$, so not QAS, and not AS.

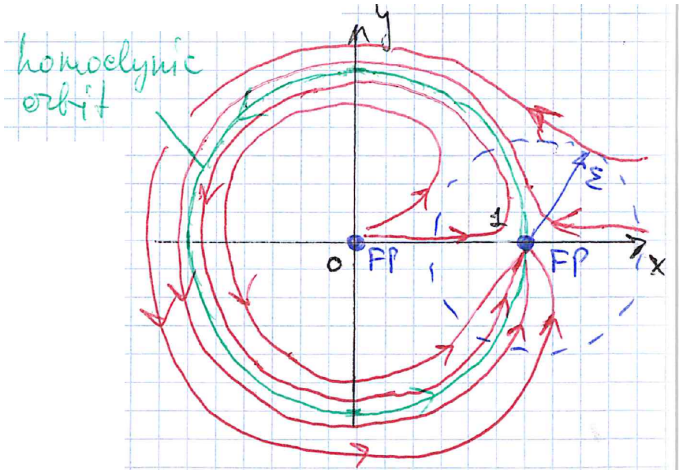


Example Λ is QAS but not LS

$$\begin{pmatrix}\dot{r}\\ \dot{\theta}\end{pmatrix} = \begin{pmatrix}r(1-r^2)\\ \sin^2\frac{\theta}{2}\end{pmatrix}$$

Linearized has one $\lambda = 0$, line of fixed points $\parallel$ to y-axis starting at $(1,0)$.

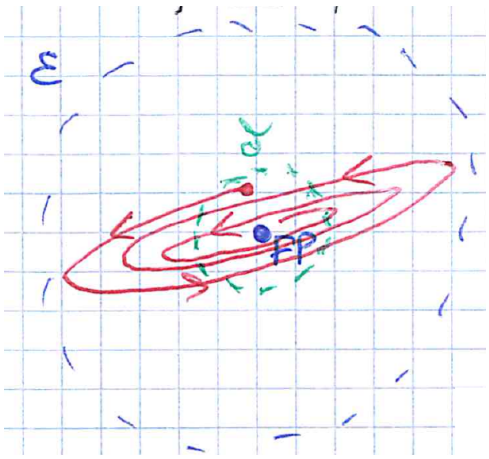
Nonlinear terms turn $(1,0)$ into a saddle-node.



For $\varepsilon < 1$ cannot find $\delta > 0$ such that trajectory is in $N_\varepsilon((1,0))$ — it will leave it. So, not LS! But is QAS as all trajectories eventually converge at FP $(1,0)$. So, not AS.

Example AS with non-monotonic convergence

Stable focus, elongated in shape.



LS since for any $\varepsilon > 0$ can always find $\delta > 0$ such that $\phi_t(\bar{x})$ with $\bar{x} \in N_\delta(\bar{0})$ always fits in $N_\varepsilon(\bar{0})$. Also QAS, obviously, and thus AS.

But approaches FP non-monotonically!

Definition Basin of attraction

If Λ is an asymptotically stable invariant set, its basin of attraction is

$$B(\Lambda) = \{\bar{x} : \phi_t(\bar{x}) \to \Lambda \text{ as } t \to \infty\}.$$

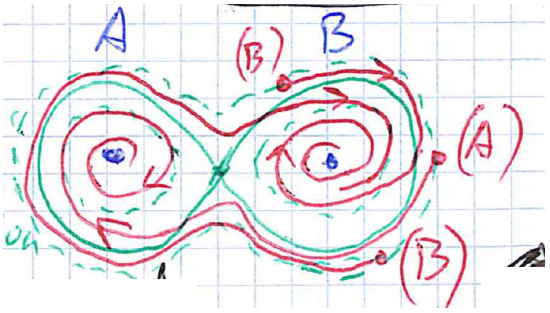
If $B(\Lambda) = \mathbb{R}^n$ then Λ is globally attracting (or globally stable).

Example In last two examples FPs were globally attracting (with some caveats in the saddle-node case).

Example Basin of stability can be non-trivial.

E.g. double potential well with weak friction.

Hamiltonian contours w/o friction



For isolated FPs can use linearized system in some cases.

Theorem Stability of hyperbolic fixed points.

If $\bar{x}_0$ is a hyperbolic stable focus or stable node then it is asymptotically stable (all hyperbolic sinks are AS). If $\bar{x}_0$ is a hyperbolic fixed point with at least one $\mathrm{Re}(\lambda_i) > 0$, then it is unstable.

§3.2 Lyapunov functions

We can prove important results about the stability of fixed points of a given flow $\phi_t(\bar{x})$ if we can find some functions that behave in a certain way under the flow.

Definition A continuously differentiable (C^1) function $V(\bar{x}) : \mathbb{R}^n \to \mathbb{R}$ is a Lyapunov function for a flow $\phi_t(\bar{x})$ on a domain D containing the origin (fixed point) if

(i) $V(\bar{0}) = 0$ and $V(\bar{x}) > 0$ for $\bar{x} \neq \bar{0}$ in D and

(ii) $\dot{V} = \bar{f} \cdot \nabla V \leq 0$ in D (or $V(\phi_t(\bar{x})) \leq V(\bar{x})$ for $\forall\, t \geq 0$).

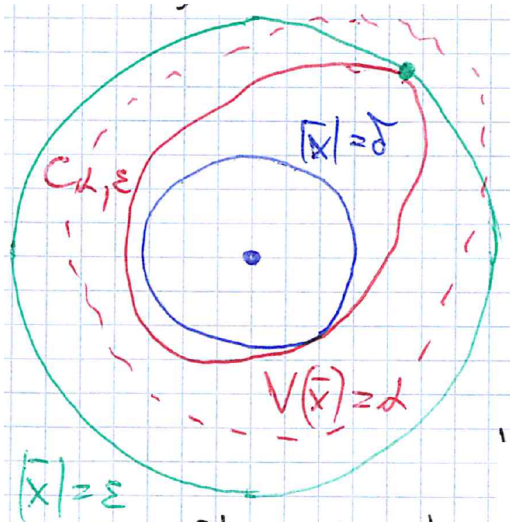
Then we have these 2 theorems.

Theorem Lyapunov's 1st theorem

Suppose that a dynamical system $\dot{\bar{x}} = \bar{f}(\bar{x})$ has a fixed point at the origin. If a Lyapunov function exists, then the origin is Lyapunov stable.

Proof Let's take small enough $\varepsilon > 0$ such that $\{|\bar{x}| < \varepsilon\} \subset D$. Let

$$\alpha = \min\big(V(\bar{x}),\ |\bar{x}| = \varepsilon\big) > 0 \quad (V(\bar{x}) = 0 \text{ only at the origin!}).$$



Now choose δ:

$$\{|\bar{x}| < \delta\} \subset C_{\alpha,\varepsilon}$$
$$= \{\bar{x} : V(\bar{x}) < \alpha,\ |\bar{x}| < \varepsilon\},$$

then

$$(|\bar{x}| < \delta) \subset C_{\alpha,\varepsilon} \subseteq \{|\bar{x}| < \varepsilon\}.$$

Then if $\bar{x} \in \{|\bar{x}| < \delta\}$, we have

$$V(\phi_t(\bar{x})) \leq V(\bar{x}) < \alpha,$$
$$\Rightarrow \quad \underline{\phi_t(\bar{x}) \in \{|\bar{x}| < \varepsilon\}}.$$

Thus, for $\forall\, \varepsilon > 0$ found $\delta > 0$: $\phi_t(\bar{x}) \in N_\varepsilon(\bar{0})$ if $\bar{x} \in N_\delta(\bar{0})$.

End of Lecture 8